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Nested-Separator Contact Closure and the Physical Response Cocycle

Research draft for the marked-width MLC program

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Abstract

For the binary direct-decoration pullback system, we prove that the complete all-depth physical contact history is a Markov state carried by the current nested puzzle boundary. Nonconsecutive shell bypasses are recovered exactly when the accumulated prefix is glued to the next shell; no all-pairs historical contact graph and no no-bypass assumption are needed. After normalization by the actual univalent branches, the structural contact state is a finite mask with an explicit recursion. We construct the exact prefix-to-cell Dirichlet response recursion, extend it to every already frozen finite protected-collar inventory, and retain all word-dependent presentation maps in co-moving coordinates. The resulting contact-aware response cocycle exposes the next major obstruction: universal response/Value compactness and a strict mixed transition. These results do not prove MBTT, UPTCR, or MLC.

1 Direct-decoration system

Let P be the direct-decoration puzzle piece, let $g = f^q$, and use only the binary univalent pullbacks

$$Q_L, Q_R \subset P, \qquad g : Q_a \to P, \quad a \in \{L, R\}.$$

Write $\phi_a = (g|_{Q_a})^{-1}$. For a word $w = a_1 \cdots a_m$, put

$$D_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}(P), \qquad h_w = g^m|_{D_w} : D_w \to P.$$

For one finite address prefix, abbreviate $D_j = D_{a_1 \cdots a_j}$ and $D_0 = P$. Then

$$D_0 \supset D_1 \supset \cdots \supset D_H \tag{1.1}$$

is a nested chain of closed Jordan puzzle disks with finite ray/equipotential boundary graphs. No one-arc boundary-intersection hypothesis is imposed.

2 Nested-separator Markov property

Define

$$\Omega_j = \operatorname{int} D_0 \setminus D_j, \qquad S_{j+1} = \operatorname{int} D_j \setminus D_{j+1}. \tag{2.1}$$

These open sets may be disconnected.

Theorem 2.1 (Nested-separator contact theorem)**.** *For every $j < H$:*
1. $\Omega_j \cap S_{j+1} = \varnothing$*;*
2.

$$\overline{\Omega_j} \cap \overline{S_{j+1}} \subset \partial D_j; \tag{2.2}$$

3. away from finitely many puzzle vertices, the active interface is

$$I_j = (\partial D_j \cap \operatorname{int} D_0) \setminus D_{j+1}; \tag{2.3}$$

4. gluing equal Sobolev traces on I_j reconstructs $\Omega_{j+1} = \operatorname{int} D_0 \setminus D_{j+1}$ and adds Dirichlet energies;
5. every contact of $\overline{\Omega_j}$ with any future shell lies on ∂D_j.

Proof. The two open sets lie on opposite sides of the closed disk D_j. A common closure point is therefore both in D_j and in the closure of its complement, which proves (2.2). At a nonvertex point of ∂D_j, the prefix approaches exactly when the point is interior to D_0, while the next shell approaches exactly when the point is not in the closed child D_{j+1}. This is (2.3). On each piecewise analytic interface arc, the two sets occupy opposite sides. The standard H^1 patching theorem identifies the Sobolev space on the union with pairs having equal traces, and energy is additive. Every later shell is contained in D_j, proving the final assertion. □

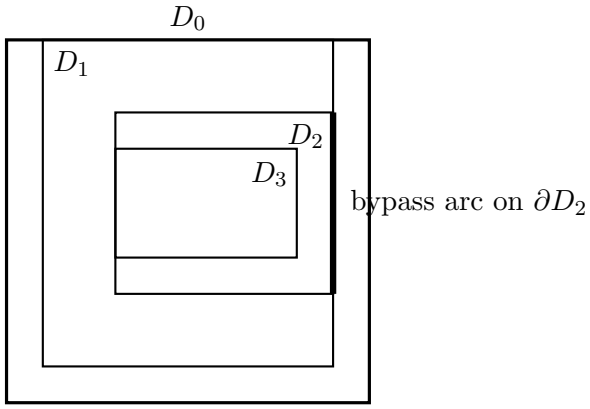


Figure 1: The audit's alternating-attachment bypass. When D_3 is added, $D_0 \setminus D_2$ is the accumulated prefix, and the entire bypass is an ordinary part of the current separator ∂D_2.

Corollary 2.2 (All-depth bypass capture)**.** *Every nonconsecutive shell contact is captured at the unique stage at which it becomes a prefix–next-shell contact. Historical shell-pair data are not needed for exact response composition.*

3 Frozen protected collars and obstacles

Let $Z \subset \overline{D_0}$ be a finite horizon-frozen union of regular protected collar tails, obstacle walls, ideal-interval walls, and electrode arcs. Put

$$\Omega_j^Z = (\operatorname{int} D_0 \setminus D_j) \setminus Z, \qquad S_{j+1}^Z = (\operatorname{int} D_j \setminus D_{j+1}) \setminus Z.$$

Theorem 3.1 (Frozen-collar separator calculus)**.** *Every common contact of $\overline{\Omega_j^Z}$ and $\overline{S_{j+1}^Z}$ lies on ∂D_j. The genuine glue is the part of I_j not declared forbidden by Z. Every boundary component of Z is retained with its declared side-port or boundary-condition type.*

Proof. Deleting Z can remove approach regions but cannot create a point approached from both sides of the closed separator away from ∂D_j. Subdivide I_j at the finite set of intersections with Z and apply the regular interface patching theorem on the remaining arcs. □

4 Persistence and finite contact mask

Take the finite common refinement of $\bigcup_{j=0}^{H} \partial D_j$, and call a connected open arc after vertex deletion a boundary atom.

Theorem 4.1 (Persistence intervals)**.** *For a boundary atom J, the set*

$$\{j : J \subset \partial D_j\}$$

is empty or an integer interval.

Proof. If $J \subset \partial D_i \cap \partial D_k$ with $i < j < k$, then $D_k \subset D_j \subset D_i$ gives $J \subset D_j$. A point of J cannot be interior to D_j, since a small disk in $D_j \subset D_i$ would contradict membership in ∂D_i. Hence $J \subset \partial D_j$. □

Corollary 4.2. *A physical boundary port persists through consecutive levels, becomes active when the nested boundary first leaves it, and never physically reactivates.*

Now choose a finite root atomization

$$\mathcal{E} = \pi_0(\partial P \setminus V_{\mathrm{geom}}), \qquad M = |\mathcal{E}|,$$

containing all geometric contact endpoints. For $a \in \{L, R\}$, let $A_a \subset \mathcal{E}$ be the atoms shared by ∂P and ∂Q_a, and refine so that

$$\tau_a : A_a \to \mathcal{E}, \qquad \tau_a(J) = g(J), \tag{4.1}$$

is an atom map. Define

$$R_j = \{J \in \mathcal{E} : h_{w_j}^{-1}(J) \subset \partial D_0\}. \tag{4.2}$$

Theorem 4.3 (Contact-mask recursion)**.** *If $a = a_{j+1}$, then*

$$\boxed{R_{j+1} = \tau_a(R_j \cap A_a).} \tag{4.3}$$

Consequently the complete structural contact history factors through the finite automaton

$$R \mapsto T_a(R) := \tau_a(R \cap A_a), \qquad R \in 2^{\mathcal{E}}.$$

Proof. A root-shared point of ∂D_{j+1} lies, by persistence, on ∂D_j. Its h_{w_j}-image is simultaneously in R_j and on ∂Q_a, hence in $R_j \cap A_a$; its new label is obtained by applying g. The converse follows by pulling an atom of $R_j \cap A_a$ back through h_{w_j}. □

5 Four-way interface classification

For a state R and symbol a, define

$$\begin{aligned} G(R,a) &= (\mathcal{E} \setminus R) \cap (\mathcal{E} \setminus A_a), \\ T(R,a) &= (\mathcal{E} \setminus R) \cap A_a, \\ O(R,a) &= R \cap (\mathcal{E} \setminus A_a), \\ Z(R,a) &= R \cap A_a. \end{aligned} \tag{5.1}$$

Class	Physical status	Operation
G	prefix and cell both approach	glue equal traces and eliminate
T	prefix approaches, cell absent	zero-energy transport by τ_a
O	prefix absent, cell approaches	couple to fixed root boundary
Z	both absent	retain in new root-shared mask

A finite protected overlay further subdivides these atoms but does not create a fifth structural approach type.

6 Exact response recursion

For a finite regular active boundary union B, use the intrinsic quotient trace space

$$\mathcal{T}_\Omega(B) = H^1(\Omega)/\{u : \operatorname{Tr}_B u = 0\} \tag{6.1}$$

with the minimum-energy quotient norm. Let q_j be the exact closed response of Ω_j^Z on fixed external terminals, the active current separator, and all proof-live side ports. Let c_a be the exact response of the fixed root cell $(\operatorname{int} P \setminus Q_a) \setminus Z_a$.

Theorem 6.1 (Geometric separator response recursion)**.** *There is an exact variational update*

$$q_{j+1} = \mathfrak{R}_{a_{j+1},R_j}(q_j) \tag{6.2}$$

that glues G-traces, transports T-traces, couples O-traces to root-side data, removes Z from the active trace, retains every proof-live side port, and minimizes precisely the internal traces. Its output is the closed Dirichlet response of the actual domain Ω_{j+1}^Z.

Proof. Transport the next shell by h_{w_j}; it is one fixed root cell. The separator theorem identifies the complete interface, and (5.1) gives all approach patterns. Sobolev patching imposes equality only on G. On T the same physical arc remains on the child boundary and carries no cell energy. On O only the cell approaches the fixed root boundary. On Z neither conducting side approaches. Add energies and minimize the internal traces. The resulting form is the Dirichlet form of the actual glued domain, so it is closed without invoking an abstract closed-form pushforward theorem. □

Theorem 6.2 (DtN Markov sufficiency)**.** *Two prefixes with the same current mask, external labels, separator coordinate map, and exact response form are indistinguishable by every common finite future continuation from the binary root-cell catalog.*

Proof. Every future step is the deterministic update (4.3), (6.2). Induct on the continuation length. □

Corollary 6.3 (Geometric live frontier)**.** *Future cells can meet the past only through the current separator. Its structural contact frontier has at most M atoms, independent of address depth. An old side port absent from the current separator and absent from the proof ledger is geometrically unreachable by every future cell.*

7 Co-moving response cocycle

The audit's welding objection applies to a global tail uniformization whose word-dependent endpoint maps are discarded. Here no such map is discarded. The actual conformal map $h_{w_j} : D_j \to P$ defines the inner trace coordinate at every step, and

$$h_{w_j}(S_{j+1}) = \operatorname{int} P \setminus Q_{a_{j+1}}.$$

Theorem 7.1 (Contact-aware physical response cocycle)**.** *The complete finite-horizon response is generated by the skew product*

$$(R, q) \longmapsto \big(T_a(R), \mathfrak{R}_{a,R}(q)\big), \qquad a \in \{L, R\}, \tag{7.1}$$

on one fixed root-normalized trace category. All finite-horizon contact and welding data are exact. This theorem does not assert norm precompactness or strict contraction of the response orbit.

8 Contact-neutral and conducting alternatives

Let $\mathcal{C}_{\mathrm{contact}}$ have vertex set $\mathcal{E}$ and edge $J \xrightarrow{a} \tau_a(J)$ whenever $J \in A_a$.

Theorem 8.1 (Contact transient/recurrent dichotomy)**.** *Every zero-cell-energy contact trajectory either leaves $\mathcal{C}_{\mathrm{contact}}$ before visiting more than M distinct atoms, or enters a recurrent strongly connected component that must be retained as an exact neutral presentation mode.*

For one fixed finite-dimensional proof feature space, let $G_\nu \geq 0$ be the accumulated physical-energy Gramian of a word ν of length L, and let $N_\nu = \ker G_\nu$.

Theorem 8.2 (Blockwise Gramian gap)**.** *For one fixed direct-decoration block and one fixed L, there is $\gamma_L > 0$ such that*

$$\langle G_\nu x, x\rangle \geq \gamma_L \operatorname{dist}(x, N_\nu)^2 \tag{8.1}$$

for every admissible word ν of length L.

Proof. There are finitely many words and finite-dimensional Gramians. Take the minimum of all positive eigenvalues of the nonzero Gramians. □

Warning 8.3. The constant γ_L, the feature dimension, and M are block-dependent. Uniform control over unbounded decorations is not proved.

9 Promotion boundary and next theorem

The results prove the split node

`PHYSICAL_END_SEPARATOR_CONTACT_REALIZATION`.

They also prove the exact finite-horizon response calculus for every already frozen finite protected inventory. They do not construct the universal future inventory or prove compactness of the response/Value orbit. Hence the broad collar gate, H2, MBTT, UPTCR, and MLC remain open.

The next major theorem is

`CONTACT_AWARE_RESPONSE_COCYCLE_COMPACTNESS_AND_VALUE_OBSERVABILITY.`

It must retain recurrent contact modes, prove a universal observability gap on the transient complement, factor every WAD and Value observable through a uniformly controlled quotient, compare primitive/neutral/virtual charts in one gauge, and dominate the full discounted error by a strict complete-block margin. This is the direct bridge from the physical geometry to H2 and MBTT.